SIMATS ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO1-AT3-Coding Chellenge

View Only

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QUESTIONS

Q1
Smart University Payroll Management System
25 marks

Q2
Smart Vehicle Insurance Claim System
25 marks

2/2 answered

Q2 **Smart Vehicle Insurance Claim System** 25 marks

Problem Statement An insurance company processes claims for three categories of vehicles: Car, Bike, and Truck. The claim amount depends on the vehicle type. Requirements Superclass: Vehicle Private members: vehicleNumber ownerName insuredAmount Subclasses: Car Bike Truck Each subclass overrides calculateClaimAmount() using different formulas. The application should: Read details for N vehicles Store them using an array of superclass references Calculate claim amounts using runtime polymorphism Search by vehicle number Display the vehicle with the highest claim amount Question 3 (Very Hard) Online Shopping Order Processing System (50 Marks) Problem Statement An e-commerce company sells products in three categories: Electronics Clothing Grocery The billing process varies for each category. Requirements Superclass: Product Private members: productId productName price quantity Subclasses: Electronics Clothing Grocery Each subclass overrides calculateBill(). The program should: Store products in an array of superclass references Calculate the final bill using runtime polymorphism Search products by ID Display the most expensive order Compute total revenue

Claim Formula (Recommended)

Car

Claim = insuredAmount × 0.80

Bike

Claim = insuredAmount × 0.60

Truck

Claim = insuredAmount × 0.90

Test Case 1
Input
3

Car
TN10AB1234
Ramesh
800000

Bike
TN09XY5678
Suresh
100000

Truck
TN11TR9999
Logan
1500000

Search Vehicle:
TN09XY5678
Expected Output
Vehicle : TN10AB1234
Claim Amount : 640000.0

Vehicle : TN09XY5678
Claim Amount : 60000.0

Vehicle : TN11TR9999
Claim Amount : 1350000.0

Highest Claim Vehicle
TN11TR9999

1350000.0

Vehicle Found

Owner : Suresh

Claim Amount : 60000.0

Your Code

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args
        Scanner sc = new Scanner(System.i

        int n = Integer.parseInt(sc.nextI
        Vehicle[] v = new Vehicle[n];

        for (int i = 0; i < n; i++) {
            String type = sc.nextLine().t
            while (type.isEmpty()) type =

            String number = sc.nextLine()
            String owner = sc.nextLine().
            double amount = Double.parseD

            if (type.equalsIgnoreCase("Ca
                v[i] = new Car(number, ow
            else if (type.equalsIgnoreCas
                v[i] = new Bike(number, c
            else if (type.equalsIgnoreCas
                v[i] = new Truck(number,
        }

        String temp = sc.nextLine().trim(
        while (temp.isEmpty()) temp = sc.

        String search;
        if (temp.equalsIgnoreCase("Search
            search = sc.nextLine().trim()
        else
            search = temp;

        Vehicle highest = v[0];
        for (Vehicle x : v) {
            System.out.println("Vehicle :
            System.out.println("Claim Amc
            System.out.println();
```

```java

                if (x.calculateClaimAmount()
                    highest = x;
                }
            }

            System.out.println("Highest Claim
            System.out.println(highest.getVeh
            System.out.println(highest.calcul
            System.out.println();

            boolean found = false;
            for (Vehicle x : v) {
                if (x.getVehicleNumber().equa
                    found = true;
                    System.out.println("Vehic
                    System.out.println("Owner
                    System.out.println("Claim
                    break;
                }
            }

            if (!found) {
                System.out.println("Vehicle r
            }

            sc.close();
        }
}

class Vehicle {
    private String vehicleNumber;
    private String ownerName;
    private double insuredAmount;

    Vehicle(String vehicleNumber, String
        this.vehicleNumber = vehicleNumbe
        this.ownerName = ownerName;
        this.insuredAmount = insuredAmour
    }

    public String getVehicleNumber() {
        return vehicleNumber;
    }

    public String getOwnerName() {
        return ownerName;
    }
```

```

    public double getInsuredAmount() {
        return insuredAmount;
    }

    public double calculateClaimAmount()
        return insuredAmount;
    }
}

class Car extends Vehicle {
    Car(String v, String o, double a) {
        super(v, o, a);
    }

    public double calculateClaimAmount()
        return getInsuredAmount() * 0.80;
    }
}

class Bike extends Vehicle {
    Bike(String v, String o, double a) {
        super(v, o, a);
    }

    public double calculateClaimAmount()
        return getInsuredAmount() * 0.60;
    }
}

class Truck extends Vehicle {
    Truck(String v, String o, double a) {
        super(v, o, a);
    }

    public double calculateClaimAmount()
        return getInsuredAmount() * 0.90;
    }
}
```

Input

```
3
Car
TN10AB1234
Ramesh
```

Output

```
Vehicle : TN10AB1234
Claim Amount : 640000.0

Vehicle : TN09XY5678
Claim Amount : 60000.0
```

Submitted